



ANDRI GILANG APKARA

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A Physics undergraduate specializing in Instrumentation and Medical Physics, with hands-on experience in medical equipment calibration and technical support Internship, laboratory and lecturer assistance, and IoT Project Builder. Possesses strong analytical thinking, technical adaptability, and leadership skills developed through academic, organizational, and internship experiences.

Work Experiences

BRIN – Pusat Riset Teknologi Kelistrikan

Feb 2026 - Present

MBKM Intern – Microgrid Research Group

- Developed PV power forecasting system by integrating real-time weather data via BMKG API for improved energy prediction
- Implemented Modbus RTU and TCP/IP protocols for data acquisition and communication between microgrid devices
- Designed and built a mini SCADA system using Flask for real-time PV monitoring and visualization

PT Quantum Inti Akurasi - Jakarta, Indonesia

Jun 2025 - Aug 2025

Intern – Radiology, Radiotherapy (Medical Physics) & Instrumentation

PT Quantum Inti Akurasi specializes in calibration, maintenance, and testing of medical equipment/industry, across radiology and radiotherapy departments

- Completed a 2-month rotation in Radiology, Radiotherapy (Medical Physics), and Instrumentation departments.
- Assisted in calibration, maintenance, troubleshooting, and daily quality assurance of medical equipment including CT-Scan machines.
- Developed and delivered a final project titled "IoT-Based Smart Monitoring System for CT-Scan Rooms", enabling real-time temperature monitoring to protect sensitive devices.
- The project was positively evaluated by the Director and management for its innovation and practical impact.

MAN 8 Jakarta – Robotics Club - Jakarta, Indonesia

Aug 2023 - Aug 2024

Robotics Trainer & Team Coach

- Coached a 10–15 member robotics team in programming, algorithm design, and mechanical/embedded systems, developing structured weekly training modules.
- Successfully led the team to win multiple national-level competitions by supervising research, prototype development, testing, and final performance preparation.

Teaching & Laboratory Instruction Experience - Depok, Indonesia

Feb 2024 - Dec 2025

Teaching Assistant (Fundamental Physics I) & Laboratory Assistant (Basic Physics Laboratory)

- Competitively selected as 1 of only 5 out of ~130 students to serve as Teaching Assistant in Semester 3, and later chosen as 1 of 6 laboratory assistants from 20–30 applicants in Semester 5.
- Facilitated tutorials and laboratory sessions for 35–50 students, preparing equipment, guiding practical experiments, reinforcing safety procedures, and supporting student learning through progress monitoring and structured academic assistance.

Freelance – Private Tutoring - Jakarta, Indonesia

Aug 2023 - Aug 2024

Private Tutor (Physics & Mathematics)

- Delivered personalized tutoring in physics and mathematics, focusing on exam preparation, problem-solving techniques, and scientific reasoning.
- Improved students' academic performance and confidence through tailored lesson plans and continuous progress evaluation.

Education Level

University of Indonesia - Depok, Indonesia

Aug 2023 - Aug 2027 (Expected)

Bachelor of Bachelor of Science in Physics with specialization in Medical and Instrumentation Physics, 3.35/4.00

- Active member of HMD Fisika UI; later appointed as Head of the Social Department and contributed to improving departmental engagement and outreach. Earned competitive selections as both Teaching Assistant and Laboratory Assistant during undergraduate studies.

Jakarta State Polytechnic - Jakarta, Indonesia

Aug 2022 - Aug 2023

Bachelor of Applied Engineering (D4 Energy Conversion Engineering Technology), 3.50/4.00

- Completed first-year studies focused on mechanical systems and energy conversion engineering.
- Discontinued after Semester 2 due to a motorbike accident, with all completed courses passed successfully.

Organisational Experience

Physics Student Association – University of Indonesia (HMD Fisika UI) - Depok, Indonesia <i>Head of Social and Enviroment Department</i> HMD Fisika UI is a university-level student organization that facilitates academic, social, and community engagement programs for Physics students at the University of Indonesia.	Nov 2024 - Present
- Appointed as Head of the Social & Environment Department after serving in the division, demonstrating strong leadership in redesigning content strategies to increase student engagement. - Transformed previously low-engagement Instagram article posts into short, simple, and easy-to-read formats, and introduced creative and humorous reels to improve accessibility and interest. - Successfully generated positive feedback from the student community and increased content reach and engagement, contributing to a measurable improvement in departmental performance.	
OSIS/MPK MAN 8 Jakarta - Jakarta, Indonesia <i>Secretary to the Student Council President & Student Council Leadership Candidate</i> OSIS/MPK is the main student leadership organization responsible for managing school-wide events, student affairs, and organizational governance.	Jul 2020 - Jun 2021
Served as Secretary to the Student Council President after being selected as one of the top three candidates for Student Council President, winning the second-highest vote count. Assisted in administrative coordination, document management, event planning, and communication between divisions to support student governance activities.	
Research Club (KIR MAN 8 Jakarta) - Jakarta, Indonesia <i>Research Club Leader</i>	Jan 2020 - Jun 2021
- Led the school's Research Club by coordinating student research projects, mentoring members, and supervising competition preparation. - Organized study groups and workshops to improve research-writing and scientific presentation skills among members.	
Research Club (KIR Se-Jakarta Timur) - Jakarta, Indonesia <i>Litbang</i>	Jan 2019 - Jun 2020
- Represented the school at the East Jakarta Research Division (Litbang), participating in regional-level scientific activities and research discussions. - Contributed to coordination between schools, supported event organization, and assisted in evaluating student project submissions at the regional level.	

Projects, Awards & Achievements

- Project Manager – MUNAS IHAMAFI XVIII (National Level) (2025):** Led the national organizing team for the 18th National Congress of Indonesian Physics Student Associations, coordinating delegates across Indonesia and managing strategic planning, decisions, timelines, and execution at a national scale.
- IoT-Based Smart Monitoring System for CT-Scan Rooms (2025):** Developed a real-time temperature monitoring system using IoT sensors to maintain optimal CT-Scan operating conditions during internship at PT Quantum Inti Akurasi.
- Earthquake Detection System – Mechanical Sensor + Transistor Flip-Flop Alarm (2023):** Developed a practical final project for Electronics I by designing an earthquake detection device using a mechanical vibration sensor integrated with a transistor-based flip-flop (BJT-NPN) alarm system. The circuit successfully triggered stable visual/audio alerts upon detecting mechanical disturbances. Awarded Best Final Project among peers for functionality, circuit stability, and innovative design.
- National University Physics Olympiad - UNM (NuPhO) (2019):** Reached the Semifinalist stage in the NuPhO organized by Universitas Negeri Malang, competing against university-level participants as a high school representative.
- National Science Olympiad - Puspresnas (KSN) – Physics (2020):** Represented the school at the City/Regency level of the National Science Olympiad (KSN), participating in advanced rounds of the national physics competition.
- Madrasah Science Competition - Kemenag RI (KSM) – Physics (2020):** Won 2nd Place in the Provincial Level Madrasah Science Competition (KSM) DKI Jakarta, representing the school in the regional physics competition.
- Nusantara Robot Contest – Robot Creative Category (2019):** Won the Bronze Prize in the Robot Creative Category at the Nusantara Robot Contest (UIN Jakarta & IYRA), competing at the national level.
- Jakarta Robotic Competition - DKI Jakarta Deputy Governor’s Cup (Maze Solving Senior Category) (2021):** Achieved Second Place in the Maze Solving Senior Category at the Jakarta Robotic Competition, held by Dunia Robot Indonesia and supported by the DKI Jakarta Deputy Governor’s Cup.
- Information Technology Academic & Certification Excellence (2019):** Recognized as the Best Graduate in Information Technology and achieved the second-highest IT certification score at SMPIT Gema Nurani, demonstrating strong academic performance and solid foundational skills in computer operations and digital technology.